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Changing opinion, knowledge, skill and behaviour of Vietnamese shrimp farmers by using serious board games

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ABSTRACT

Purpose: Mekong Delta's shrimp farming contributes to socio-economic development but tends to reduce the mangrove area. On the one hand, NGOs advocate balancing ecology and economy, while on the other hand, the Vietnamese government supports intensifying shrimp production. The latter strategy increases shrimp diseases and marginalises smallholders. To influence the opinions, knowledge and behaviour of shrimp farmers, we developed and tested a serious board game.

Approach: Through several iterations, we designed a realistic board game mimicking local shrimp farming. Then, after conducting three sessions of gameplay with farmers in three sites, we assessed how this gameplay influenced players. We used a semi-experimental set-up by collecting data on the day of the gameplay, and on two further time-points.

Findings: After the gameplays, farmers reported to have learned about the risks of monoculture-intensive-shrimp system and the advantages of hybrid systems. Likert ratings showed that players adopted innovations faster and, in case of shrimp diseases, consulted commercial service-providers less, but with other farmers more. They confirmed to have learned that the best strategy for them would be to adopt hybrid systems, i.e. intensify part of the area of mixed-mangrove or improved-extensive shrimp farms.

Theoretical implications: Realistic board games can create opportunities for social learning and training of groups which are not always easy to reach.

Practical implications: Board games can support transfer of knowledge and skills to farmers, and can develop their critical thinking.

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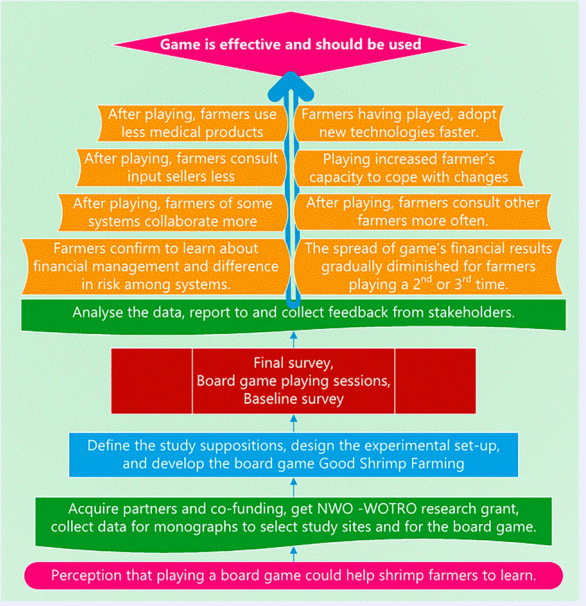
Coastal zone; mangrove; learning; training; extension; communication

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Creation: The findings that board games are effective in the context of a top-down extension systems offer value opportunities.



Introduction

Coastal environments are always evolving (Walther et al. 2002) but continuous and increasing anthropogenic activities, such as shrimp aquaculture (Thomas et al. 2017), make people the major threat for mangrove forests. Widespread intensification of shrimp culture may lead to increased risks among others from diseases (Joffre, Poortvliet, and Klerkx 2018; Shinn et al. 2018), related bankruptcies (Binh et al. 2005) and decreased overall shrimp yield (Joffre et al. 2015). Likewise, the non-accounted harmful environmental impacts (Primavera 1991) increased inequality (Adger 1999) and income-decimating diseases (Shinn et al. 2018) have mostly resulted in a negative long-term economic impact of shrimp aquaculture (Rivera-Ferre 2009).

Bush et al. (2010) describe two scenarios for sustainable shrimp production: intensive bio-secure monoculture and mangrove landscape integrated systems. The integrated systems, such as shrimp sylvo-aquaculture (Bosma et al. 2014), respect the standards of certification systems such as ASC (Aquaculture Stewardship Council), which requires shrimp farms to have at least 50% of mangrove cover within or around the farms. One of the systems complying with this standard is the Mixed-Mangrove Shrimp Systems (MMS) practiced by the farmers in the Mekong delta's semi-protected zone, between 300 and 600 metre from the coastline (Ha et al. 2013).

Environmental non-government organisations (NGOs) advocate the expansion of these climate-smart MMS farms to support a more balanced development of ecology and economy (Fabres 2011). However, goals and strategies in both private and public sectors are working against this. In particular, the Vietnamese government aims to increase shrimp yield, economic turn-over, food security and export by providing

subventions and training for intensification and processing. Moreover, farmers are hard to convince to plant mangrove forest because they want to intensify shrimp production as copy-cat behaviour (Goldstone and Janssen 2005). They also fear losing mandate over the land once trees are planted (Bosma et al. 2014), they lack land and capital or they need to recover former investments in intensive shrimp farming (Joffre et al. 2015).

As for technology transfer, the Vietnamese extension services practice top-down knowledge transfer (Schad et al. 2011) by using training, demonstration and contact-farmers, i.e. farmers who, assisted by the extension services, apply technology and on whose farms others are invited for lessons. Top-down approaches may reach transfer of knowledge, but create information gaps (Brown, Llewellyn, and Nuberg 2018) and deter critical thinking and social learning. Lack of critical thinking leads to undesired copy-cat behaviour, while social learning leads to improved agricultural extension impact (Yishay and Mobarak 2017).

To encourage critical thinking among the Vietnam shrimp farmers, we piloted a serious board game, based on the theoretical framework described by Rodela, Ligtenberg, and Bosma (2019). Serious games are recognised as an educational tool that can transfer knowledge (Hu et al. 2018; Wait and Frazer 2018), favour social learning; and are useful in particular when behaviour can be mimicked through demonstration (Smith and Berge 2009; Bandura 1977).

We supposed that playing the board game would (1) improve individual shrimp farmers' knowledge on financial risks and advantages of the different production systems, (2) increase the sharing of knowledge and contribute to social-learning, and finally (3) change farmer's behaviour. In the long term, increased awareness on the risks of the intensive monoculture system might enhance their willingness to maintain, or shift to more sustainable practices, such as MMS, but the project period was too short to measure this.

This article reports on the longitudinal study that assessed whether a board game could indeed contribute to changes in shrimp farmers' opinion, knowledge and behaviour. In the next section, Material and Methods, we present the study sites, the game, the variables, samples and data collection, and the data-analysis for the three types of data collected. Thereafter, we present, in four sections, the outcomes regarding the farmers' opinion, knowledge, skills and behaviour. We then discuss the results and finally formulate our conclusions.

Material and methods

To test whether playing the board game could change shrimp farmer's opinion, as a first step, we collected related data across three study sites. Then we defined the variables of interest including knowledge, attitude and behaviour related to learning, the four shrimp systems, risk and environment. This paper doesn't describe the iterative process through which we developed a realistic game that could change these variables. During that process, some farmers played a pilot version without technology cards (Rodela, Ligtenberg, and Bosma 2019).

Shrimp and mangrove in the study sites

Four main types of shrimp production systems can be distinguished (Table 1): MMS, improved extensive (IE), intensive culture of *Penaeus monodon* (IMO) and intensive

Table 1. Main characteristics of the shrimp farming systems in the two provinces in 2016.

	Farm size	Crop cycles	Stocking density	Shrimp yield**	Operation cost	Gross income	Start-up Cost***	Net income	Mangrove pond ratio
System	ha	n yr^{-1}	numbers m^2	kg ha^{-1} yr^{-1}	USD ha^{-1} yr^{-1}	USD ha^{-1} yr^{-1}	USD ha^{-1}	USD ha^{-1} yr^{-1}	%
MMS*	4.3	1.0	4	490	550	13,600	2,600	5,200	31
IE	3.1	1.5	7	450	1,300	21,700	3,000	3,500	6
IMO	2.2	2	25	6,150	23,800	50,200	11,900	11,900	0
IVA	2.5	4	80	12,500	31,100	54,000	13,800	15,700	0

Notes: Legend: MMS: Mixed-Mangrove Shrimp farms; IE: Improved extensive; IMO: intensive culture of *P. monodon*; IVA: Intensive culture of *L. vannamei*.
*MMS have more products; Total income is about double the one mentioned here.
**Total shrimp yields for one year.
***Not equal to annual fixed cost.

culture of *Litopenaeus vannamei* (IVA). We selected three project sites in two provinces in the north-west of the Mekong delta (Figure 1), where all four shrimp production systems are practised and having issues on mangrove conservation. Two sites were located in Duyen Hai district of Tra Vinh province: the first site covered the communes of Long Vinh and Long Khanh; the second, the communes of Long Toan and Truong Long Hoa. The third site covered the three coastal communes of Binh Dai district in Ben Tre province. In both provinces, intensive shrimp culture is the focus of local authorities and farmers; and most mangrove areas are managed by forestry management services or semi-public companies.

Site 1 includes more than 50% of the forest coverage of Duyen Hai district. There, local government invested in electricity and roads to facilitate intensified shrimp culture;

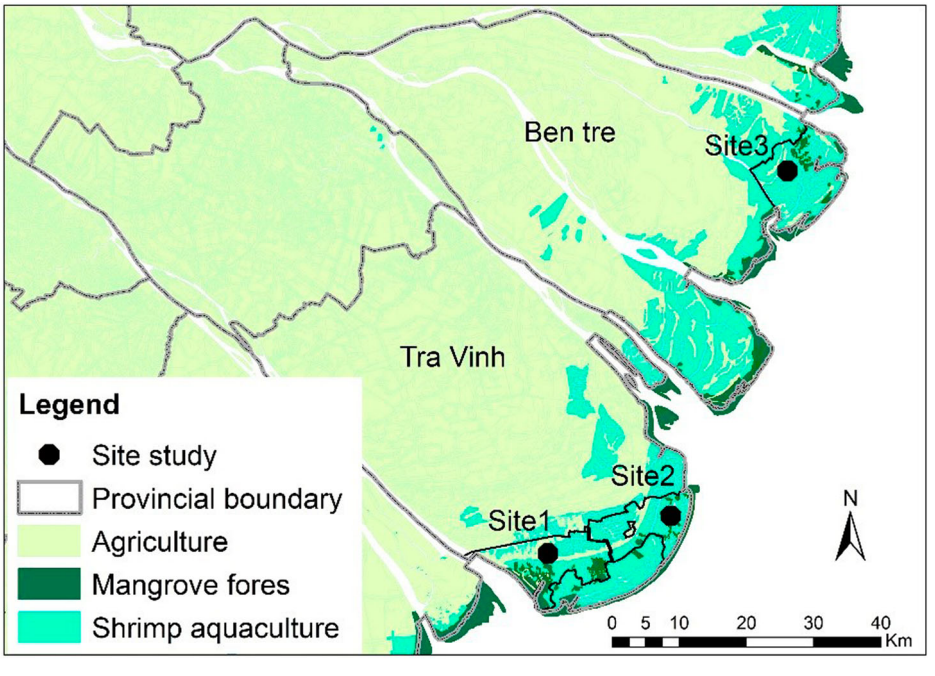


Figure 1. Land use in the coastal area and the location of the study sites.

however, plans to develop mangroves for tourism failed due to disagreement among departments and lack of investment. In site 2, the local government had a stronger focus on intensive shrimp culture and a lenient attitude toward mangrove forest. Site 3, located in Binh Dai district of Ben Tre province, had its local government aiming to increase intensive shrimp farming and allowing farmers to intensify MMS, while maintaining the tree cover. The district has the largest area of shrimp ponds, but the smallest mangrove cover among the province's coastal districts.

In general, MMS farms are larger than 4 ha; IE farms cover about 3 ha, while the average IMO and IVO farms are about 2,3ha (Table 1). The stocking densities and yields of MMS and IE remain low. The stocking density of IVA per cycle is 3 times higher than that of the IMO. The net margin of IVA (15,700 USD ha^{-1} yr^{-1}) is just about 20% higher than that of the IMO (11.900 USD ha^{-1} yr^{-1}) because IVA sells lighter and thus cheaper shrimp. In the year of the survey (2016), the average farm gate prices of shrimp were approximately 6.90 (MMS), 6.70 (IE), 6.50 (IMO) and 6.20 (IVA) USD kg^{-1}; prices are different as the shrimp price is determined by size: the more shrimp in one kg, the lower the price is. The yields of MMS and IE are lower than those of the IMO and IVA; but as MMS and IE have low operational cost and less disease impact, the net margin of the MMS (5.200 USD ha^{-1} yr^{-1}) is about 1/3, and that of the IE (3,500 USD ha^{-1} yr^{-1}) is about ¼ of the IVA. The farms in all shrimp production systems have about 2.2 full-time workers on average.

The serious board game

In 2016 and 2017, we conceptualized a realistic and challenging serious board game (Rodela, Ligtenberg, and Bosma 2019). The game was expected to modify farmer's perception of risk on shrimp disease outbreaks and on the importance of mangrove. To make the game realistic, we included two types of risk cards, general and personal. Each type carried a positive or negative impact on the player's availability of cash, a negative impact on the disease risk, and an option to buy risk reduction technologies. The technology options included, next to training, among others, nets protecting the pond from birds, plastic lining, filter pond and membership of a farmer's group. The board had 15 farm plots differing in size and access to water, roads and electricity (Supplement 2). The price of the plots with good infrastructure (access to electricity, road and water) was about 3.3 times higher than that with poor infrastructure (no electricity, no road). The game can be played by up to six players, whose turn to play is determined by throwing dices. Thereafter they choose one plot, on which all start playing with an extensive system and an amount of cash corresponding to the plot size. Thereafter, they play eight rounds either sun- or clockwise and at their turn, take a personal card. All rounds start with taking a general risk card. From round two of the game, the players may invest in either MMS, IMO or IVA, on part or of their entire plot; from round five they may invest in technology by buying the related cards.

Although the cropping cycle of the systems is different, to make the game not too complex, the game assumes that the farming cycle of every system is equal (6 month); the average cost and profits are adjusted to this period. Through field work and expert workshops, we collected the related data (Table 1) that were rounded to simplify calculations in the game, while to create contrast, we chose some amounts on either left or

Table 2. Cost and benefits per 6-month cycle of the four shrimp farming systems implemented in the board game (Million Vietnamese Dong ha^{-1}).

	MMS	Improved Extensive	*Monodon* (IMO)	*Vannamei* (IVA)
Investment cost	90	100	150	150
Operational cost	90	20	250	400
Minimum & maximum gross margin for the dice-scores	+20/+130	−50/+150	−250/+400	−400/+450

right side of the distribution (Table 2). The disease risk, impacting the gross margin per round, is determined in each round by the score of the two 10-sided dices (one for the tens and one for the units). This score can be modified either negatively or positively by one of the risk cards, or positively by investing in technologies. The gross margin is read from Table 2, in five ranges of the final calculated score.

Variables, sample and data collection

To test the supposition, we monitored two groups of shrimp farmers: non-players and players of the board game. Before playing the game and some months after, we collected data on shrimp farmer's opinion on issues relating to shrimp farms, mangrove and environment. We also collected data on their behaviour regarding risk, learning, and how fast they take up or adopt a new technology on which they were informed, had seen, or otherwise learned about (Supplement 1). Answers on 23 questions were graded on a seven-step-Likert scale. Two questions had an open sub-question asking for an explanation, and the last was a choice question on *Actions taken*, with six close- and one open-ended question.

For these surveys, we selected 219 shrimp farmers from among the four systems and in about equal numbers from the three study sites, in 2016. One month after all game playing sessions, out of 219 farmers, only 173 of them were interviewed again; and only those data from farmers who were interviewed twice were considered in the analysis of this longitudinal survey. Among these, 54 didn't play the game and served as a control group. Among the farmers who were interviewed twice, 43 practised IE; 111, MMS; 86, IVA and 19, IMO. This total is higher than 173 because of the hybrid systems, i.e. IE and MMS farmers were also simultaneously practicing IVA or IMO. Only one farmer doing only IMO had played the game, thus the answers of players and non-players from this system were not compared on itself, but only considered in the overall analysis.

We made some farmers play the eight rounds of the game once and others twice at several months' interval in 2017 and 2018. At communal level, farmers were invited to play for a one whole morning game session where they played in groups of 4 to 6, at 2 or 3 tables. Some of the invited farmers, selected among the 219 to play the game, didn't show up. About 40 more farmers played the game to fill up at least two tables during a session. We recorded the game results from all players. Among the 119 players, 5, 42 and 110 had participated at three, two and one sessions, respectively. Of these 110 players, 46 played the pilot version of the game. All 157 who had played the game were asked one open question 'What did you learn from this game?' and eight questions with a 4-step-Likert scale (Exit questionnaire). The selection of farmers was partly decided by the extension services. The researchers provided a list of farmers to invite and a minimum number of participants. If the latter was not reached, the extension

agents invited other farmers from their network. We can't exclude a bias, but this bias would concern in particular the 40 complementary invited farmers who were not considered in the longitudinal survey.

Data analysis

All collected data were transcribed in MS-Excel®. The detailed analyses of the three types of data were done as described below.

Gameplay outcomes and exit questionnaires

The outcomes of playing the game were analysed in SPSS® with a focus on the player's systems choices and the monetary results at the end. The rankings in the exit questionnaire were interpreted from the frequency distributions in SPSS®, but similar descriptive answers were counted in MS®-Excel.

Longitudinal survey

As a first step for the analyses of the Likert questions in SPSS®, we checked the normality of the data. Then we compared the means for both groups, players and non-players, with a T-test for the effect of time (survey 1 before, versus survey 2 after), and the effect of having played the game (1) or not (0). If the *T*-test did not confirm an effect of time for the non-players, the effect of time among the players was analysed by using an ANOVA. While interpreting the results of the T-test, we considered the Equality of Variance. To compare the ratings given by the farmers from the four systems, we made cross tabs with the frequency graphs.

The choices on the answers of the last question, *Actions taken*, were calculated as percentage of all answers given by the farmers. Multiple answers were transposed in two other columns. Through the option '7', *Other actions*, many farmers gave multiple answers of which most aligned with one or more of the 1st six options; in the latter case the '7' was replaced with that option, and the other options were added in a 2nd or 3rd column of the spreadsheet. Based on the answers on *Other actions*, two options were added: *Do nothing* and *Change water*, and these two options were added in the frequency distribution.

Results

The farmers and other stakeholders judged the game as realistic. Almost all farmers found the game easy to play; they were satisfied with their own performance, and could use their real life's know-how during the game. We found that playing either the final version of the game or the pilot (without the event and technology cards), did not influence significantly the means of the ratings given by these groups. The number of times the farmers played the game did not also significantly influence the means of the ratings in the exit questionnaires.

Skills on financial management

The spread in the monetary outcomes of the game diminished for the farmers participating in a second or third session (Figure 2). By playing the game, the farmers learned to avoid financial losses, while also the number of players with exceptionally high gains

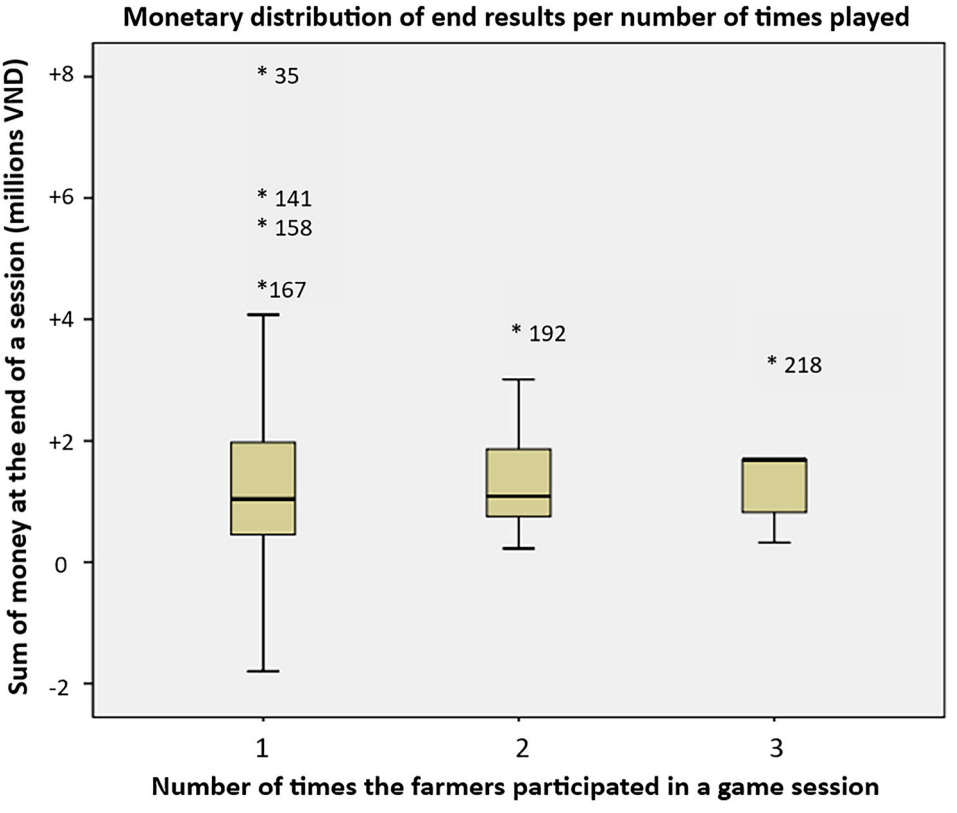


Figure 2. The distribution of the monetary gains after the eight rounds for farmers having played the game once (1.0), twice (2.0) or thrice (3.0), for all locations.

diminished. The main strategy to avoid losses due to the risks of intensive shrimp monoculture was to shift only part of their farm to intensive shrimp ponds, while maintaining either IE or MMS in the other pond areas. The general trend was that a higher percentage of players were doing an intensive system on (part of) their plots.

Knowledge on economic risks

In the exit questionnaire, 90 to 100% of the players confirmed to have improved their knowledge on the economic risks of shrimp farming. They said that by playing the board game, they learned about: investment & operation costs, other methods to farm shrimp, measures and technologies to improve their shrimp farming. Moreover, they said to have learned that MMS give sustainable profits, while intensive systems equate to higher levels of risk, and that a shift to a suitable system increases income.

Opinions

To make the ratings more objective during the longitudinal survey, we asked farmer's opinions on Income, Risk and Risk of Disease by rating first their own system and then the other systems with reference to their own system (Supplement 1).

Opinions on shrimp farming system's income and risks

Both player' and non-players' rating on income stability of a hybrid of MMS with IE decreased (4.4 to 3.3) over time. Likewise, farmers of IE, IVA and IMO judged the income stability of a hybrid MMS with either IMO or IVA about the same, irrespective of their having played the game or not (Supplement 3). In contrast, the control MMS farmers themselves ranked this income stability higher the 2nd time (Figure 3(a); mean Likert 3.2 to 4.4, i.e. very low to low), while the MMS players didn't change their ranking (Figure 3(b): 3.6 and 3.5). Note that, results for the control which showed low rankings (Figure 3(a)), and that for the players which showed high rankings (Figure 3 (b)) converged to the center. In the 2nd survey, the control IVA farmers ranked the profit of their IVA farm significantly lower ($p < 0.01$) than that of the other shrimp systems. However, the IVA players gave the same ranking during the 1st and 2nd survey.

At the start, the opinion on the risk of disease in MMS significantly differed between the players and the control group ($p < 0.05$). After having played the game, the players' rating of the risk shifted significantly ($p < 0.05$) upward for IE (from 3.1 to 3.6) and for MMS (2.8 to 3.5; Supplement 4). After having played the game, the IVA farmers appreciated the risk higher in their own system (3.3 to 4; Figure 4(c)), as well as in the IE system (Figure 4(a)). The non-playing IVA farmers hardly raised their rating of the risk in their own systems (Figure 4(c)). After having played the game, the IVA farmers appreciated the risks in both their own and the IE system higher than before (Figure 4(a)). After having played the game, the IE farmers, and to a lesser extent the MMS farmers, found that the risk in IVA and IMO were lower. The different tendencies explain that this overall rating did not change significantly among the players.

Opinions on risk of disease and role of mangrove

The means of both non-players and players indicate that there was no shift in the disease frequency in their own system: MMS (from 3.1 to 3.4), IE (3.2 to 3.5), IMO (3.5 to 3.7) and IVA (3.6 to 3.7). However, when judging also the other systems, both non-players and players found that the disease frequency increased in MMS (from 3 to 5) and IE (3.5 to 5), i.e. from every 5 years to every 3 years, while this increase was smaller for IMO and IVA (4.4 to 5.3). The assumptions of homogeneity of variance and normality were violated for these rankings (Supplement 4).

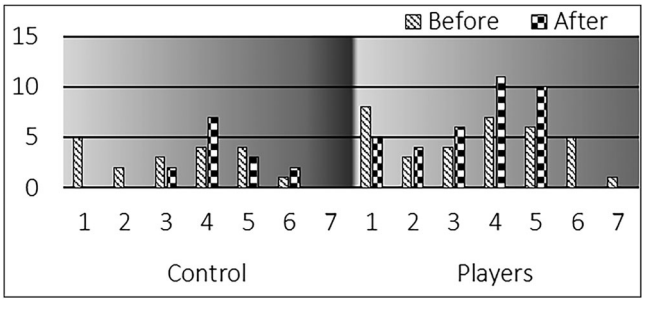


Figure 3. The number of non-playing (3a = left) and playing (3b = right) MMS farmers who rated, on a scale of 1 to 7, the income stability of the hybrid of MMS with either IMO or IVA, at the start (1) and the end of the study (2).

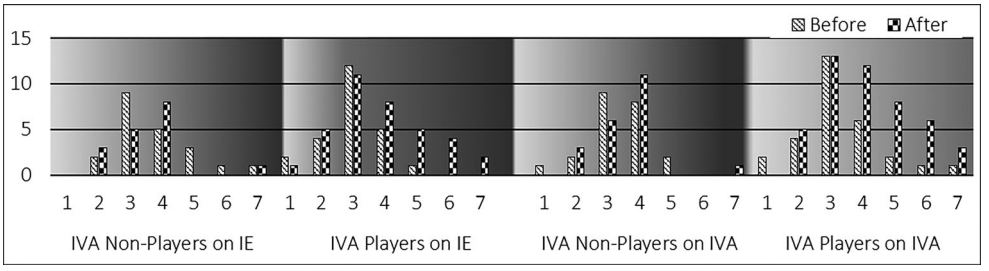


Figure 4. The number of IVA farmers who rated, on a scale of 1 to 7, the frequency of diseases in the IE system (4a = left = Non-players; 4b = centre-left = Players) and in their own system (4c = centre right = Non-players; 4d = right = Players), at the start (1) and the end of the study (2).

The ranking by both non-players and players on the role of MMS and mangroves for shoreline safety decreased over time ($p < 0.001$). Both groups attributed a higher importance to mangrove than to MMS for shrimp disease prevention, and this did not change in time (Supplement 5). The narrative explanations of both their ratings tended to be similar with those within their communities.

Opinions on influence by others and from training

Playing the game did not change the mean rating on the influence others might have on the practice at their farm (Supplement 6). However, this rating and the one on the utility of training (see hereafter) were not normally distributed, and the assumptions of homogeneity and equality of variance were violated; thus no conclusions on significance could be made. The number of non-players who had never assisted a training at the start was higher (15 on 53) than that of the players (8/117); at the end, the non-participation of the non-players had diminished: 9/54. While on average the utility attributed to these trainings was not affected by playing the game, the MMS and IVA farmers playing the game tended to have a more positive appreciation of the usefulness of these trainings (Figure 5(a–d)).

Behaviour on consultation and adoption

The number of the players' informal meetings about shrimp farming with other farmers in the hamlet increased after the game ($p < 0.05$). The number of people (in village and beyond) who asked for advice about shrimp farming also increased slightly ($p < 0.1$), in particular for the farmers of the IE system (Figure 6).

The interviewed farmers' tendency to share their know-how with other farmers on dealing with shrimp diseases on their farm increased over time ($p < 0.001$) from occasionally (3.2) to frequently (4.7). Overall, there was no significant impact of playing the game (Figure 7(ab)), but among IVA farmers the tendency increased from sometimes (3.2) to frequently (5.3) (Figure 7(cd)).

The farmers' actions, which they said to have taken whenever one of their ponds was struck by a disease outbreak, changed over time; some of these changes seemed independent of playing the game (Table 3). The farmers' most frequently used strategy remained: harvest, sell and prepare the pond for a new cycle, sometimes after a rest; the non-players

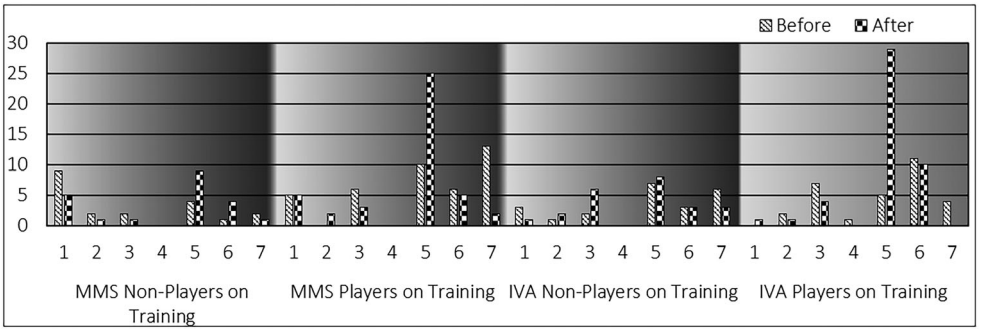


Figure 5. The rating given by MMS farmers (Non-players = 5a at left; Players = 5b at centre-left), on a scale of 1 to 7, on the usefulness of training by a public body, and by IVA farmers (Non-players = 5c = centre-right; Players = 5d at right), at the start (1) and the end of the study (2).

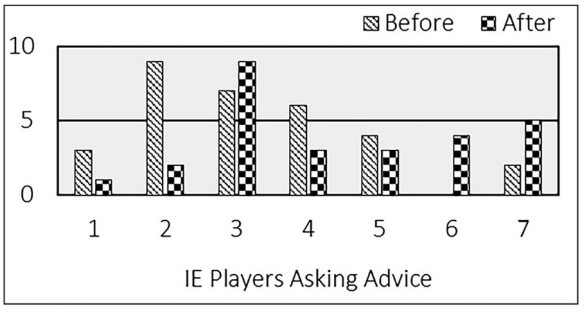


Figure 6. The number of people (in village and beyond) whom the IE farmers asked for advice about shrimp farming, at the start (1) and the end of the study (2).

did this less but the players more, after two years. Overall, more farmers checked whether colleagues had similar problem; their use of medicinal products diminished by 5%, and their frequency of consultations with veterinary pharmacies decreased from 20% to 4%. The non-players consulted extension agents more, and also changed more often to another system.

The ratings on the speed of adoption of technologies proposed at trainings were not normally distributed, but homogeneity and equality of variance were not violated. Players stated to have adopted new technology faster after a training than non-players ($p < 0.1$), but within the group of players, both IVA and MMS farmers notably increased their adoption of new technology (Figure 8(b)).

Discussion

Parts of the study's sample are small, the groups are not well-balanced and other external factors (projects meetings) may have impacted differently on the groups. The last two factors seem inevitable in field research. We obtained larger group sizes by merging the dataset of the participants of the 'pilot' stage, after verifying whether the means of their rating were statistically equal to that of the other interviewed farmers. Future studies

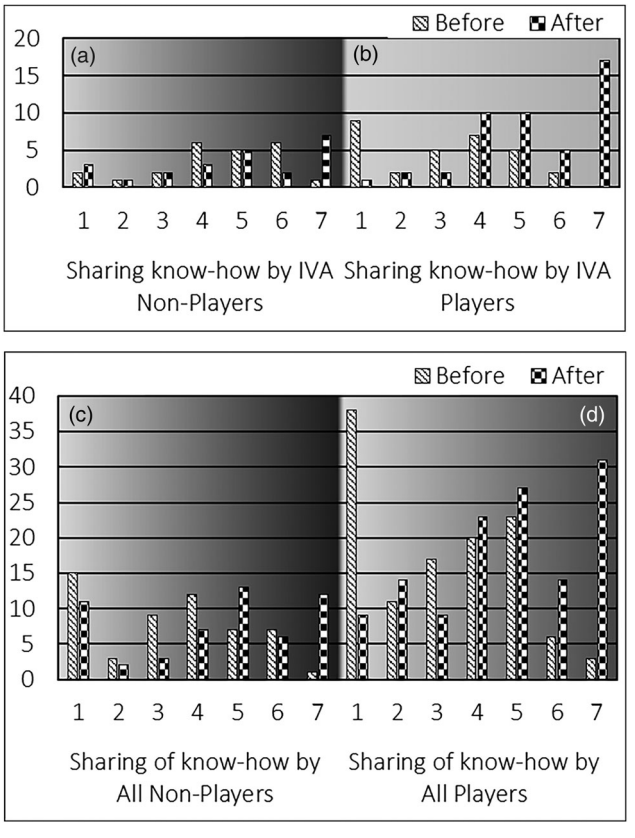


Figure 7. (ab) Changes in the frequency of sharing know-how with other farmers when dealing with shrimp diseases as given by all non-players (7a, left) and all players (7b, right), at the start (1) and the end of the study (2). (cd) Changes in the frequency of sharing know-how with other farmers when dealing with shrimp diseases as given by the non-playing (7c, left) and playing IVA farmers (7d, right), at the start (1) and the end of the study (2).

Table 3. The frequencies of the actions farmers (total % from 205 answers by 174 farmers) said to have in case of a disease outbreak, before and after playing (Yes) or not playing (No) the game.

	Before			After		
Answers of the farmers	Total	No	Yes	Total	No	Yes
Do nothing	5.9	4	7	0.0	0	0
Harvest and start new cycle	44.4	51	39	44.5	40	49
Buy drugs	23.4	26	21	32.8	36	30
Ask if others have similar problem	5.4	5	6	9.7	9	10
Ask extension	3.9	1	6	6.1	7	6
As veterinary pharmacy	10.2	6	14	2.0	2	2
Change system	1.0	0	2	3.6	6	2
Change water	5.4	6	5	1.2	2	1

should increase sample size to cover for the drop-outs among the sampled farmers and avoid the 'pilot' stage.

Our control group (non-players) was drawn from the same list of the players; irrespective of being control or player, all farmers meet each other at social/family events and most

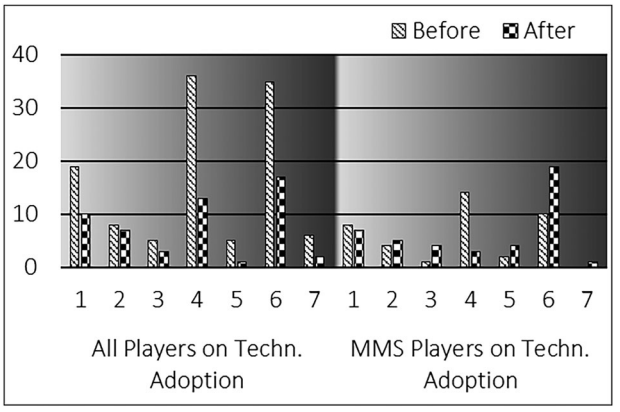


Figure 8. The effect of playing the game on rating the speed of adopting new technology after training by all farmers (6a, left) and the MMS farmers (6b, right), at the start (1) and the end of the study (2).

may meet at (coffee)shops (although some MMS farmers live in relative isolation). Nevertheless, the control had less contact with other farmers than the players had due to the board game sessions; thus, this meeting on itself may already have stimulated them to talk more on the topic. To prevent this, future research could consider providing a control training activity.

Know-how and opinion on shrimp farming

The farmers' opinions after having played the game, together with the reduced spread in the financial outcomes of the plays, indicate that the game can be used as a tool for learning on the (financial) management of shrimp farms. This finding is corroborated by Hu et al. (2018) on students at a medical faculty and with Wait and Frazer (2018) on employers in the retail industry. The gameplay confirmed this trend towards intensification, and farmers who played the board game learned that practicing a hybrid system, i.e. intensifying only part of their IE or MMS reduced their risk. At present the risk management strategies of shrimp farmers are limited (Joffre, Poortvliet, and Klerkx 2018).

There was a discrepancy in the general opinion on the frequency of diseases in shrimp farming. Over time the frequency wouldn't have changed within their own farms (all means were between once every 4–5 years), but when judging the other systems, both non-players and players found that the disease frequency increased in MMS and IE from once every 5 to every 3 years; and in IMO and IVA, from once every 4 to every 3 years. As mentioned, just before the study began, the new income-decimating shrimp disease, AHPND, (Shinn et al. 2018) was frequently mentioned in newspapers and news on TV or radio. The bad news on the sector might have influenced opinions, while most individuals within the sample didn't suffer from an increased impact.

The narrative explanations by the interviewed farmers on their rating of the influence of mangrove and MMS on disease prevention tended to converge within the communities, i.e. within one community, farmers gave about the same reasons. Other communities tended to give another set of explanations; the sample in the three sites covered in total

seven communities. Such finding seems to indicate that the difference lies in sources or gaps in communication/information/knowledge.

Behaviour on consultation and adoption

We found three shifts in behaviour among farmers who had played the game: speed of new technology adoption, frequency of meeting other farmers and actions taken in case of shrimp disease. The self-reported answers in the exit questionnaire and the speed of adoption of new technology, in particular for MMS and IVA farmers, indicate that a board game is an effective tool to increase farmers' awareness on the fit of technology for their farm. The relatively speedy adoption among MMS farmers might be related to the low frequency of their contacts with the extension services due to the longer travel distance and more costly means of transport to reach the training locations (Ha et al. 2013). Their invitation to play the game, or to participate in a survey, may have been their first entrance to a network, as shown by the diminished no-training-participation of the non-players. The larger increase among IVA farmers might be related to an urgent need for new technologies, such as filter ponds, to reduce impact of the new disease AHPND (Shinn et al. 2018) and several risk-reduction technologies were included as options in the board game. The significant reduction in consultations with the veterinarians, aligns with the slightly reduced use of drugs. The former is compensated by higher frequencies of consultations among farmers when they observe disease in their shrimp.

In particular, the IMO and IVA farmers in the sample might have been individuals who were better connected to the extension services and had participated in training more frequently. In the Vietnamese context of top-down knowledge transfer, the use of group-learning methods is confronted with several constraints (Schad et al. 2011), and information gap is one of the barriers to innovation (Brown, Llewellyn, and Nuberg 2018). The board game, therefore, can be a tool to stimulate communication among farmers and enhance social learning which is an efficient tool to improve the impact of agricultural extension (Yishay and Mobarak 2017).

Theoretical and practical implications

The top-down agricultural extension practice that is common in Vietnam (Schad et al. 2011) has led to negative perceptions about these public services (Ngomane 2010). Focusing more on the transfer of technological solutions to clients/farmers, among others, this practice overlooked the heterogeneity of its target group, and advanced problem modes of thinking instead of engaging the end-users in the pursuit of their aspirations. Ngomane (2010) called for appreciative inquiry approaches that support individuals and communities to identify and release their potential for socio-economic advancement. Our board game not only can be used by extension agents to share information about risk and the medium- and long-term performance of different shrimp farming systems, but can also be used to increase critical thinking and social capital. Farmers who players this board game reduced their number of consultations with product sellers, but increased their interaction times and information exchange with community members after gameplay. Although the potential of board games for social learning and socio-economic advancement needs to be assessed case by case (Rodela, Ligtenberg, and Bosma 2019),

this paper presents a case of a game that can be used as a tool by extension services to engage farmers on a given subject matter in a more realistic way.

While the effectiveness of the top-down extension method 'Contact-Farmer', increased through 'Training', both hardly reached other farmers from outside designated core areas (Kondylis, Mueller, and Zhu 2017). However, since exchanging information with and among peers increases technology adoption, (Yishay and Mobarak 2017) stimulating this information exchange or social learning is an efficient communication strategy of extension services (Tjernström 2017). Our findings confirm that board games can play a practical role in such strategies. Although the affordability of developing and using a game may be disputed (Ditzler et al. 2018), to improve sustainability of shrimp farming a good approach to learning and social learning might be as effective as (or increase the impact of) clustering (Joffre, Poortvliet, and Klerkx 2019), and more efficient than aerial-based approaches requiring costly changes in the infrastructure for water supply and drainage (Bottema, Bush, and Oosterveer 2018).

Government support for intensification may lead to increased bankruptcies and thus marginalise up to 5% of the farmers (Ligtenberg et al. 2018; Joffre et al. 2015). This study also confirmed that farmers who played the board game practiced less risky hybrid systems. If government aligns its policy with this farmers' strategy, the board game-based learning tool can then be used to support extension services, along with other tools (Misiko and Halm 2016) to reduce exclusion and marginalisation (Rice et al. 2019), and improve farmer's risk management skills (Joffre, Poortvliet, and Klerkx 2018).

Conclusions

Playing the board game allowed farmers to explore the (long-term) impact of their decisions in a safe environment related to the risks and opportunities of various production strategies, and thus created space for learning. Moreover, after playing the game, the farmers increased information exchange among their peers, and consequently, increased the likelihood of technology adoption. Thus using the board-game can support the effectiveness of top-down extension programmes for shrimp farmers. Our results indicate that the game should be played at least two times at some interval. The cost in terms of time for the extension services will be rather limited while the benefits are clear. The main drawback remains the cost and time for developing realistic games.

Policies on shrimp farming and practices of shrimp farmers tend to intensify the production systems. However, by playing the board game, farmers realise that hybrid forms, i.e. including an intensive pond in a mixed-mangrove shrimp farm can reduce risks, which, according to explorative models, is also valid in larger scale. Thus, using board games as an educational and training tool can strengthen policies that aim at balancing economic and ecological goals to reduce on the one hand risks and impacts of diseases breakouts, and on the other hand exclusion and marginalisation of smallholders in the shrimp sector.

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